

7/22/23, 4:03 AM

Codio - Image Composition and Blending

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imageGen.py

55

56from PIL import Image

57

58# Generate two images using the DALL-E 2 API

59image1_url = generate_base_image('forest')

60image2_url = generate_base_image('sunset')

61

62

63# Download and open the generated images

64image1 = download_image(image1_url, 'image1')

65image2 = download_image(image2_url, 'image2')

66

67# Create a custom mask (grayscale gradient)

68width, height = image1.size

69mask = Image.new('L', (width, height))

70for y in range(height):

71 for x in range(width):

72 mask.putpixel((x, y), x)

73

74# Composite the two images using the custom mask

75result = Image.composite(image1, image2, mask)

76

77# Save the composited image to a file

78result.save('composite_custom_mask.jpg')

79

80# Blend the images using the Multiply mode

60% (60:1)

Terminal

Welcome to Ubuntu 18.04.5 LTS (GNU/Linux 5.19.0-1028-aws x86_64)

* Documentation: <https://help.ubuntu.com>

* Management: <https://landscape.canonical.com>

* Support: <https://ubuntu.com/advantage>

* Canonical Livepatch is available for installation.

- Reduce system reboots and improve kernel security. Activate at: <https://ubuntu.com/livepatch>

*

* Welcome to the Codio Terminal!

*

Guide

Collapse

<>⚙️☰

Image Blending with Different Modes

```
from PIL import ImageChops

def average_blend(image1, image2):
    return ImageChops.add(image1, image2, scale=0.5)

# Use the custom blend mode to blend two images
image1 = Image.open('image1.jpg')
image2 = Image.open('image2.jpg')
average_blend_result = average_blend(image1, image2)
average_blend_result.save('average_blend.jpg')
```

TRY IT!

Traceback (most recent call last):

File "imageGen.py", line 59, in <module>

image1_url = generate_base_image('forest')

File "imageGen.py", line 14, in

generate_base_image

response = openai.Image.create(

File

"/home/codio/anaconda3/lib/python3.8/site-packages/openai/api_resources/image.py", line 36, in create

response, _, api_key = requestor.request(

File

"/home/codio/anaconda3/lib/python3.8/site-packages/openai/api_requestor.py", line 226, in request

resp, got_stream =

self._interpret_response(result, stream)

File

"/home/codio/anaconda3/lib/python3.8/site-packages/openai/api_requestor.py", line 619, in _interpret_response

self._interpret_response_line(

File

"/home/codio/anaconda3/lib/python3.8/site-packages/openai/api_requestor.py", line 679, in _interpret_response_line

raise self.handle_error_response(

openai.error.RateLimitError: Rate limit exceeded for images per minute in organization org-uWGK1xe2Mi8p10GD5wRZJmYi. Limit: 5/1min. Current: 6/1min. Please visit <https://help.openai.com/en/articles/6696591> to learn how to increase your rate limit.

Click here to refresh your average blend image

Creating custom blend modes can be a powerful way to achieve a unique look and feel for your image compositions. Experimenting with different blending functions can lead to interesting and unexpected results, allowing you to further enhance your compositions and unlock even more creative possibilities with the DALL-E 2 API and PIL.

Which blending mode preserves the highlights and shadows of the bottom image when combining two images?

☒ Overlay

☐ Normal

☐ Multiply

☐ Screen

The Overlay blending mode combines Multiply and Screen modes, preserving the highlights and shadows of the bottom image when the two images are combined.

Check It!

Next

https://codio.com/sandipan-dey/image-composition-and-blending/tree/imageGen.py1/1